

Monomial $f(q) = a \cdot q^5$ Report

Polynomial

$$f(q) = aq^5$$

Naive

$$g_{\text{naive}}(X) = X^5 a$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 120a \frac{\Delta^4}{\epsilon} q + 20a \frac{\Delta^2}{\epsilon} q^3 + aq^5$$

$$\begin{aligned} \text{Var}(g_{\text{naive}}(X)) &= 3628800a^2 \frac{\Delta^{10}}{\epsilon} + 1800000a^2 \frac{\Delta^8}{\epsilon} q^2 \\ &\quad + 146400a^2 \frac{\Delta^6}{\epsilon} q^4 + 4400a^2 \frac{\Delta^4}{\epsilon} q^6 + 50a^2 \frac{\Delta^2}{\epsilon} q^8 \end{aligned}$$

$$\begin{aligned} \text{MSE}(g_{\text{naive}}) &= 3628800a^2 \frac{\Delta^{10}}{\epsilon} + 1814400a^2 \frac{\Delta^8}{\epsilon} q^2 \\ &\quad + 151200a^2 \frac{\Delta^6}{\epsilon} q^4 + 4800a^2 \frac{\Delta^4}{\epsilon} q^6 + 50a^2 \frac{\Delta^2}{\epsilon} q^8 \end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = 120a \frac{\Delta^4}{\epsilon} q + 20a \frac{\Delta^2}{\epsilon} q^3$$

Unbiased

$$g_{\text{unb}}(X) = X^5 a - 20X^3 a \frac{\Delta^2}{\epsilon}$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^5$$

$$\begin{aligned} \text{Var}(g_{\text{unb}}(X)) &= 2304000a^2 \frac{\Delta^{10}}{\epsilon} + 1152000a^2 \frac{\Delta^8}{\epsilon} q^2 \\ &\quad + 96000a^2 \frac{\Delta^6}{\epsilon} q^4 + 3200a^2 \frac{\Delta^4}{\epsilon} q^6 + 50a^2 \frac{\Delta^2}{\epsilon} q^8 \end{aligned}$$

$$\begin{aligned} \text{MSE}(g_{\text{unb}}) &= 2304000a^2 \frac{\Delta^{10}}{\epsilon} + 1152000a^2 \frac{\Delta^8}{\epsilon} q^2 \\ &\quad + 96000a^2 \frac{\Delta^6}{\epsilon} q^4 + 3200a^2 \frac{\Delta^4}{\epsilon} q^6 + 50a^2 \frac{\Delta^2}{\epsilon} q^8 \end{aligned}$$

Comparison

$$\text{mean gap} = -120a \frac{\Delta^4}{\epsilon} q - 20a \frac{\Delta^2}{\epsilon} q^3$$

$$\text{variance gap} = -1324800a^2 \frac{\Delta^{10}}{\epsilon} - 648000a^2 \frac{\Delta^8}{\epsilon} q^2 - 50400a^2 \frac{\Delta^6}{\epsilon} q^4 - 1200a^2 \frac{\Delta^4}{\epsilon} q^6$$

$$\text{MSE gap} = -1324800a^2 \frac{\Delta^{10}}{\epsilon} - 662400a^2 \frac{\Delta^8}{\epsilon} q^2 - 55200a^2 \frac{\Delta^6}{\epsilon} q^4 - 1600a^2 \frac{\Delta^4}{\epsilon} q^6$$

$$\text{Bias}(g_{\text{naive}})^2 = 14400a^2 \frac{\Delta^8}{\epsilon} q^2 + 4800a^2 \frac{\Delta^6}{\epsilon} q^4 + 400a^2 \frac{\Delta^4}{\epsilon} q^6$$